Molecular Matter

Exam code: 0625 & 0972

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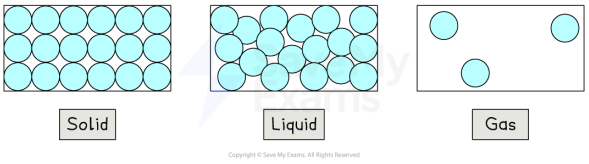
Updated on 3 February 2026

Arrangement & motion of particles

- All molecules and matter are in motion at room temperature
  - The motion and arrangement of particles must be known for each state of matter
- In a **solid**:
  - The molecules are very **close** together and arranged in a **regular** pattern
  - The molecules **vibrate** about fixed positions
- In a **liquid**:
  - The molecules are still **close** together (no gaps) but are no longer arranged in a regular pattern
  - The molecules are able to **slide past each other**
- In a **gas**:
  - The molecules are **widely separated** - about 10 times further apart in each direction
  - The molecules move about **randomly at high speeds**



Solid, liquid and gas particle model diagrams

Properties of states of matter

State	Solid	Liquid	Gas
Density	High	Medium	Low
Arrangement of particles	Regular pattern	Randomly arranged	Randomly arranged

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

	position		
Energy of particles	Low energy	Greater energy	Highest energy

The forces & distances between molecules

Extended tier only

Intermolecular forces and motion of particles

- The **forces between molecules and matter** (or 'particles') affect the state of matter
- This is because the magnitude of the forces affects the **relative distances** and **motion** of the particles
- This affects the ability of the substance to
  - Change shape
  - Change volume
  - Flow

- The word **particles** can refer to:
  - Atoms
  - Molecules
  - Ions
  - Electrons

Solids

- The molecules in a solid are held in place by **strong intermolecular** forces
  - They only **vibrate in position**
  - The **distance** between them is **fixed** and is very small
  - This gives the solid its **rigid shape** and **fixed volume**

Liquids

- The molecules in a liquid have enough **energy** to overcome the forces between them
- They are still held **close together**
  - The volume of the liquid is the same as the volume of the solid

- Molecules can **move around** (by sliding past each other)
  - This allows the liquid to **change shape** and **flow**

Gases

- The molecules in a gas have **more energy** and move **randomly** at **high speeds**
  - The molecules have **overcome** the forces holding them close together

- Because of the **large spaces** between the molecules
  - The gas can easily be **compressed** and is also able to **expand**

Worked Example

Two states of matter are described below. Identify each of the states of matter.

Substance 1

- molecules are spaced very far apart
- molecules move very quickly at random
- molecules move in a straight line

Substance 2

- molecules are quite closely packed together
- molecules move about at random
- molecules do not have fixed positions

Answer:

Substance 1

Step 1: Identify the distances between the molecules

- The molecules are spaced far apart
  - This can only describe a gas

Step 2: Identify the motion of the molecules

- The molecules move quickly, at random and in a straight line
  - This confirms that substance 1 is a gas

Substance 2

Step 1: Identify the distances between the molecules

- The molecules are closely packed
  - This could describe either a solid or a liquid

Step 2: Identify the motion of the molecules

- The molecules move at random and do not have fixed positions
  - This confirms that substance 2 is a liquid

Temperature & energy of particles

- As the **temperature** of a gas increases, so does the **average speed** of particles in the gas
  - At higher temperatures, the particles have more **kinetic energy**
- The amount of pressure that a gas exerts on its container is dependent on the temperature of the gas
  - This is because particles gain **kinetic energy** as their temperature increases

cannot travel any slower than 0 m/s

- The temperature at which all particles are stationary is called **absolute zero**
  - Absolute zero has a value of −273 °C

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